

KINEMATIC, KINETIC, AND BLOOD LACTATE PROFILES OF CONTINUOUS AND INTRASET REST LOADING SCHEMES

JAMES DENTON AND JOHN B. CRONIN

New Zealand Institute of Sport and Recreation Research, Auckland University of Technology, Auckland, New Zealand

ABSTRACT. Denton, J., and J. B. Cronin, Kinematic, kinetic, and blood lactate profiles of continuous and intraset rest loading schemes. *J. Strength Cond. Res.* 20(3): 528–534. 2006.—The purpose of this study was to investigate and compare the acute kinematic, kinetic, and blood lactate responses to continuous and intraset rest loading schemes that differed in terms of rest frequency but not total rest duration. Nine male subjects performed an isoinertial bench press task (6 repetition maximum load) with a continuous, an intraset rest equated by total rest time, volume, and load (ISRV), and an intraset rest equated by total rest time and load (ISRR) loading scheme. The scheme order was assigned in a block-randomized order with a minimum of 48 hours of recovery between testing sessions. Attached to the bar of the Smith machine was a linear position transducer that measured vertical displacement with an accuracy of 0.01 cm. Displacement data was sampled at 1,000 Hz and collected by a laptop computer running custom-built data acquisition software. Finger prick blood lactate samples were taken from the nondominant hand before exercise, immediately after exercise, and 5, 15 and 30 minutes after exercise. Blood glucose samples were taken before exercise only. It was observed that manipulating the rest period, by increasing the frequency but decreasing the length of each rest period, did not significantly influence the kinematics and kinetics associated with resistance training, but did have an effect on the postexercise blood lactate response when the load, rest duration, and training volume were equated (ISRV). This finding may be of practical significance if fatigue is important in strength development or conversely if power training requires minimal fatigue. It was also observed that increasing the frequency of the rest period enabled the subjects to perform a greater number of repetitions (ISRR), resulting in significantly greater kinematics, kinetics, and blood lactate accumulation.

KEY WORDS. rest, bench press exercise, resistance training, strength, power

INTRODUCTION

Typically, strength training that aims to improve strength through increasing the cross-sectional area of the muscle (hypertrophy) has used relatively short rest periods (60–90 seconds), whereas neuronal strength and power training are typically by longer rest periods (180–300 seconds). However, the scientific rationale on which these rest periods are based remains relatively unsupported in the literature. That is, very little research has studied the effect of different rest intervals on the development of maximal strength and power. However, there is a general consensus when using traditional, or continuous, type loading schemes that reduced rest periods increase the metabolic demands and reduce the muscle's force production capability (14, 15).

While continuous loading schemes with intersets rest

periods have been traditionally prescribed during resistance exercise, intraset rest loading schemes have also been studied (16). Intraset rest training refers to training that divides a set or sets into a smaller work unit or units with more intermittent rest periods (e.g., 3 sets of 6 repetition maximum [6RM] and 180-second rest periods vs. 6 sets of 3 repetitions of 6RM load and 72-second rest periods). It has been theorized that the short rest periods within the training set allow the partial resynthesis of the intramuscular phosphocreatine stores (4), allowing greater force and power output in the later stages of the workout (16, 17). Intraset rest loading is also thought to allow athletes to increase their training volume by training at high intensities for longer durations and/or performing additional repetitions.

The resultant kinematics and kinetics associated with training (e.g., greater force and power output, greater time under tension, greater total work) is thought to be important for the development of maximal strength and power while simultaneously minimizing the performance-inhibiting effects of metabolic accumulation and substrate depletion (7, 16, 17). However, research in this area is still in its infancy, and therefore, many of the theories regarding intraset rest loading schemes are currently unsubstantiated. Therefore, the purpose of this study was to compare the acute kinematic (displacement, velocity, and acceleration), kinetic (force, impulse, work, and power), and blood lactate responses to continuous and 2 different intraset rest loading schemes. It is thought that such a study will improve our understanding of intraset rest loading and subsequent application as a resistance strength training method.

METHODS

Experimental Approach to the Problem

A crossover experimental design was used to study the acute responses to the 3 different loading schemes that differed in terms of rest frequency but not in total rest duration. The loading schemes were assigned to the participants through a block randomization design to negate any possible order effects.

Subjects

Ten healthy male subjects volunteered to participate in this study, which was evaluated and approved by the Human Subject Ethics Committee of the Auckland University of Technology. Subjects were required to have a minimum of 12 months of resistance training experience and were prescreened for any previous or current injuries and

TABLE 1. Participant descriptive characteristics.

Age (yr) [mean (SD)]	Height (cm) [mean (SD)]	Body mass (kg) [mean (SD)]	Bench press (6RM) [mean (SD)]
25.2 (4.5)	178.5 (4.7)	79.7 (10.8)	80.8 (26.5)

**FIGURE 1.** Modified Smith machine.

medical conditions that would contraindicate participation. Additionally, subjects signed an informed consent document before testing commenced. During data collection, 1 subject withdrew from the study for unknown reasons, leaving 9 subjects. The data from this subject were not included in any analyses. The descriptive characteristics of the subjects that completed the study are presented in Table 1.

Equipment

A counterweighted isoinertial Smith machine (Fitness Works, Auckland, New Zealand) capable of weight increments of 1.25 kg was used for all strength assessments and testing sessions (Figure 1). The Smith machine was chosen for safety, easy instrumentation, availability, and subject familiarity. A mechanical brake was used at all times to allow standardization of joint angles (range of motion) and to ensure subject safety. This brake was positioned at the lowest point of the eccentric phase and did not influence the movement patterns during testing in any way.

Attached to the bar of the Smith machine was a

linear position transducer (mean sensitivity, $0.499 \text{ mV} \cdot \text{V}^{-1} \cdot \text{mm}^{-1}$; linearity, 0.05% full scale; PA80; Unimeasure, Corvallis, OR), which measured vertical displacement with an accuracy of 0.01 cm. Displacement data were sampled at 1,000 Hz and collected by a laptop computer (Toshiba, Tokyo, Japan) running custom-built data acquisition and analysis software (LabView 6.1; National Instruments, Austin, TX).

Blood samples were analyzed through portable blood glucose (Accu-Chek; Roche, Auckland, New Zealand) and blood lactate (Lactate Pro; Arkray, Kyoto, Japan) analyzers. Blood glucose and lactate accumulations were measured and recorded to the nearest $0.1 \text{ mmol} \cdot \text{L}^{-1}$.

A portable stadiometer and digital scales (Model 770; Seca, Hamburg, Germany) were used to measure height and body weight, respectively. Height was measured and recorded to the nearest 1 cm, whereas body weight was measured and recorded to the nearest 0.1 kg.

An electronic timer was used to ensure that the correct rest periods were used for each loading scheme. The electronic timer gave an audio signal when the rest time was completed, cueing the subject to begin the next training set. The investigator also gave subjects a 15-second countdown to ensure that the subjects were properly prepared to begin at the audio cue.

Experimental Testing

During preliminary testing, subjects were tested for height, weight, and 6RM bench press strength. The subjects performed 1 testing session for each of the loading schemes. Kinematic, kinetic, and blood lactate data were collected during each of these sessions. All subjects were familiarized with the bench press exercise, loading schemes, procedures, and resistance training equipment before testing to eliminate any acute learning effects. A minimum of 48 hours of recovery was prescribed between sessions, and all sessions were performed at the same time of day to reduce the impact of any diurnal variations.

Assessment of 6 Repetition Maximum

Strength (6RM) for the Smith machine bench press exercise was assessed in accordance with the following protocol. Two warm up sets of 7 repetitions at 50 and 70% of the individual's body weight were performed, separated by 5 minutes of rest. Grip width was standardized with the hands placed on the bar shoulder width apart. A mechanical brake, discussed previously, was also used to standardize the depth of each eccentric movement and to ensure subject safety. The load was conservatively increased until the subject was able to successfully perform 6 but not 7 repetitions. Five minutes of rest was prescribed between each attempt. Subjects were instructed to perform each repetition in a controlled manner and were not allowed to pause at any time. All strength assessments were conducted by the same investigator who judged successful attempts, including full range of motion, for each individual. The reliability of these procedures was calculated over 2 nonconsecutive days (within a 7-day period) and were found to be stable (coefficient of variation [CV] = 1.5%, intraclass correlation coefficient [ICC] = 0.98).

Loading Schemes

Before each testing session, a standardized warm up including 2 warm up sets with 40 and 75% of the individ-

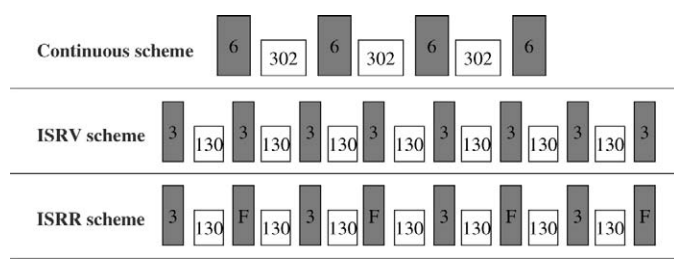


FIGURE 2. Graphical representation of the continuous and intraset rest loading schemes.

ual's previously established 6RM were performed. Five minutes of rest was prescribed between each warm-up set and between the warm-up and the first training set.

Each subject completed 3 sessions separated by a minimum of 48 hours. To avoid order effects, the scheme to be performed was assigned to each subject in a block-randomized design. Each scheme was performed with the previously established 6RM load, and subjects were instructed to perform the concentric phase of each repetition as explosively as possible.

The 3 loading schemes included a continuous loading scheme consisting of 4 sets of 6 repetitions separated by 302 seconds of rest (CONT), an intraset rest loading scheme equated by loading volume (ISRV) consisting of 8 sets of 3 repetitions separated by 130 seconds of rest, and an intraset rest loading scheme equated by total rest time (ISRR). This final scheme consisted of 8 sets of exercise; however, whereas sets 1, 3, 5, and 7 consisted of 3 repetitions, subjects were instructed to perform sets 2, 4, 6, and 8 to voluntary failure. Once more, 130 seconds of rest was prescribed between training sets. A graphical representation of the 3 schemes is shown in Figure 2.

Blood Sampling

On experimental days, blood samples were obtained from a finger of the nondominant hand using sterile techniques. Blood lactate samples ($\sim 3\mu\text{l}$) were obtained after 10 minutes of seated rest (Pre), immediately after exercise (P0), and 5 (P5), 15 (P15), and 30 minutes (P30) after exercise. Blood glucose samples ($\sim 3\mu\text{l}$) were obtained before exercise only (Pre).

Statistical Analyses

The raw displacement time data were filtered using a low-pass Butterworth filter with a cutoff frequency of 5 Hz. The filtered displacement time data were differentiated to determine velocity (displacement/time) and acceleration (velocity/time). From this data, the kinematic and kinetic variables were calculated from basic mechanical formulae. Data were calculated for the eccentric (defined as the period from maximum to minimum vertical displacement) and concentric (defined as the period from minimum to maximum vertical displacement) phases of each repetition.

Test-retest reliability for the 6RM assessment procedures was assessed using a CV ($SD/\text{mean} \times 100$) and an ICC (two-way random absolute agreement model). Eccentric and concentric time under tension, force, power, and work done was calculated for each repetition and summed to give the total for each set of each loading scheme (CONT, ISRV, and ISRR). These set totals were summed to provide eccentric, concentric, and overall totals for the

TABLE 2. Number of repetitions completed per set during the continuous and intraset rest loading schemes.*

Set	CONT [Mean (SD)]	ISRV [Mean (SD)]	ISRR [Mean (SD)]
1	6.0 (0.0)	3.0 (0.0)	3.0 (0.0)
2	6.0 (0.0)	3.0 (0.0)	7.0 (1.1)
3	6.0 (0.0)	3.0 (0.0)	3.0 (0.0)
4	5.6 (1.1)	3.0 (0.0)	5.8 (1.6)
		3.0 (0.0)	3.0 (0.0)
		3.0 (0.0)	4.4 (1.5)
Total	23.6 (1.1)	24.0 (0.0)	30.8 (7.5)

* CONT = continuous; ISRV = intraset rest equated by total rest time, volume, and load; ISRR = intraset rest equated by total rest time and load.

entire session, which were then analyzed. Impulse was calculated as the area under the force time graph. Total time under tension, force, power, impulse, and work done during each loading scheme were compared for significant differences using a one-way repeated-measures analysis of variance (ANOVA) and Holm-Sidak posthoc contrasts. Lactate responses were compared using a 2-factor (groups, 3; time, 5) repeated-measures ANOVA. Where a statistical difference or interaction was apparent, a post-hoc Bonferroni adjusted pairwise comparison was used to determine the nature of the difference. Statistical analyses were performed using SPSS version 11.5 (SPSS Inc., Chicago, IL), and the criterion level for statistical significance was set at an alpha level of $p \leq 0.05$.

RESULTS

The reliability ($CV = 1.5\%$, $ICC = 0.98$) associated with the 6RM strength assessment established during pilot testing was high, indicating the baseline maximal strength measure used in this study was stable across testing occasions. However, as can be observed in Table 2, when subjects used the ISRR loading scheme, a significantly greater number of repetitions (~ 30) were able to be performed compared with the other 2 loading schemes (~ 24). The greatest number of extra repetitions was performed earliest in the workout (set 2 = 7.0 repetitions), after which there was a steady decline.

The mean $\pm SD$ for the kinematic and kinetic variables of interest is presented in Table 3. It can be observed from Table 3 that the ISRR scheme resulted in significantly greater eccentric, concentric, and total values for all the variables studied, whereas no significant differences were observed between the CONT and ISRV schemes for any of the kinematic and kinetic variables examined. Total time under tension ($\sim 53\%$ overall), total mean force ($\sim 62\%$ overall), total impulse ($\sim 59\%$ overall), total mean power ($\sim 63\%$ overall), and total work done ($\sim 65\%$ overall) were significantly greater than the CONT and ISRV schemes. The greater kinematic and kinetic values associated with ISRR loading was to be expected because this scheme enabled more repetitions (Table 2) to be performed than the CONT and ISRV schemes.

In terms of the 2 schemes equated by volume, no significant differences were observed between eccentric ($p = 0.166$), concentric ($p = 0.655$), or total ($p = 0.974$) time under tension, despite shorter total contraction durations (~ 4 seconds) being associated with the ISRV loading

TABLE 3. Kinematic and kinetic responses to the continuous and intraset rest loading schemes.*

Variables	CONT [Mean (SD)]	ISRV [Mean (SD)]	ISRR [Mean (SD)]	F value (p value)
Eccentric phase				
Duration (s)	20.4 (3.3)	18.4 (3.2)	29.87 (6.9)†‡	8.71 (0.006)
Mean force (N)	19,028 (6,756)	19,122 (6,847)	30,769 (8,511)†‡	6.63 (0.006)
Impulse (N·s ⁻¹)	14,935 (4,135)	13,460 (3,281)	21,207 (3,698)†‡	11.6 (0.002)
Mean power (W)	6,553 (2,210)	6,701 (2,259)	11,484 (4,474)†‡	6.30 (0.007)
Work (J)	9,288 (1,356)	9,732 (1,331)	14,577 (1,834)†‡	47.7 (<0.001)
Concentric phase				
Duration (s)	33.0 (14.7)	30.1 (9.6)	58.93 (14.2)†‡	28.3 (<0.001)
Mean force (N)	19,051 (6,746)	19,165 (6,839)	30,780 (8,520)†‡	6.61 (0.006)
Impulse (N·s ⁻¹)	27,198 (14,067)	25,560 (13,253)	47,172 (16,113)†‡	37.8 (<0.001)
Mean power (W)	4,609 (1,647)	4,774 (1,140)	6,555 (1,522)†‡	4.42 (0.025)
Work (J)	9,417 (1,718)	9,769 (1,274)	14,957 (2,037)†‡	28.8 (<0.001)
Total				
Duration (s)	49.2 (11.3)	45.2 (8.2)	87.8 (16.2)†‡	20.4 (<0.001)
Mean force (N)	38,079 (13,502)	38,257 (13,685)	61,549 (17,031)†‡	6.62 (0.006)
Impulse (N·s ⁻¹)	42,423 (20,241)	37,399 (14,763)	68,103 (20,424)†‡	31.9 (<0.001)
Mean power (W)	11,161 (3,663)	11,475 (2,907)	18,039 (5,872)†‡	6.42 (0.007)
Work (J)	18,705 (3,035)	19,502 (2,757)	29,515 (3,823)†‡	37.0 (<0.001)

* CONT = continuous; ISRV = intraset rest equated by total rest time, volume, and load; ISRR = intraset rest equated by total rest time and load.

† Significantly greater than corresponding CONT value ($p \leq 0.05$).

‡ Significantly greater than corresponding ISRV value ($p \leq 0.05$).

scheme. Similarly, no significant differences were observed between eccentric ($p = 0.164$), concentric ($p = 0.814$), or total ($p = 0.362$) impulse, despite lower values ($\sim 2,000$ N·s⁻¹) being associated with the ISRV loading scheme. Also, despite subjects being asked to perform the concentric phase as explosively as possible, the eccentric phase was quicker ($\sim 39\%$) than the concentric phase during all loading schemes.

From further analysis, it seems that the ISRR loading scheme had a greater influence on the concentric time under tension ($\sim 66\%$) and impulse ($\sim 69\%$) compared with the eccentric time under tension ($\sim 34\%$) and impulse ($\sim 31\%$). Subsequently, ISRR loading had a greater influence on the eccentric mean power ($\sim 64\%$) than the concentric mean power ($\sim 36\%$). In terms of mean force and work, the ISRR loading scheme had a similar influence on the concentric and eccentric phases.

Pre-exercise blood lactate and blood glucose concentrations (blood glucose data not presented) were not significantly different between sessions (Figure 3). In terms of blood lactate response over time, the CONT and ISRR schemes resulted in significant increases in blood lactate concentrations compared with Pre, from P0 to P15, whereas the ISRV scheme resulted in significantly elevated blood lactate concentrations at P0 only. All schemes returned to Pre blood lactate concentrations by P30.

Regarding the between-group blood lactate responses, at P0 the blood lactate responses to the ISRV and CONT loading schemes were not significantly different from each other, but the ISRR scheme was significantly greater (34%) than the ISRV scheme. At P5, the ISRR scheme was observed to have a significantly greater blood lactate concentration than the ISRV (103%) and CONT (36%) schemes, whereas the CONT scheme was significantly greater (50%) than that of the ISRV scheme. At P15, the ISRR scheme was significantly greater than the CONT (30%) and ISRV (74%) schemes, whereas no significant differences were observed between the CONT and ISRV

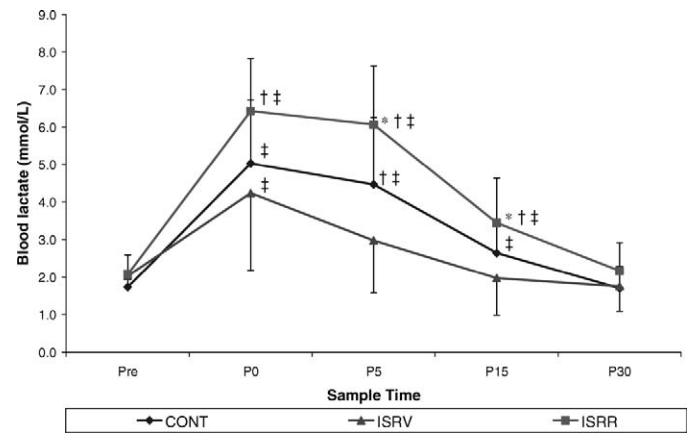


FIGURE 3. Blood lactate responses to the continuous and intraset rest loading schemes. * Significantly greater than corresponding continuous value ($p \leq 0.05$); † Significantly greater than corresponding intraset rest equated by total rest time ($p \leq 0.05$); ‡ Significantly greater than baseline value ($p \leq 0.05$).

schemes. No significant differences were observed between loading schemes at P30.

DISCUSSION

Previous comparisons of ISRV and CONT loading schemes observed significantly greater repetition power output during ISRV loading (16). However, despite using similar loading parameters (6RM), training techniques, and repetition grouping, such results were not found in this study. Instead, no significant differences were observed between the CONT and ISRV schemes in any of the kinematic or kinetic variables studied (including maximal and mean power). The blood lactate responses to the CONT and ISRV schemes did differ, however, because

lower postexercise blood lactate (P0–P15) was associated with the ISRV loading scheme. Therefore, while Lawton et al. (16) suggested that ISRV loading may provide a superior stimulus for the development of muscular power, the findings of this study did not support such that contention, but rather, we found that the ISRV loading scheme may simply offer an alternative to CONT loading schemes with lower levels of blood lactate accumulation.

Lawton et al. (16) reported significantly greater concentric time under tension during CONT loading compared with ISRV loading. Once more, despite using similar loading parameters, training techniques, and repetition grouping, these findings were dissimilar to the findings of this study. That is, no significant differences ($p > 0.05$) were observed between the CONT and ISRV loading schemes in this study. However, although not statistically significant, the total time under tension associated with the ISRV loading scheme was shorter (~ 4 seconds) than the CONT loading scheme. If time under tension is important in the development of maximal strength and hypertrophy, as previously suggested by Bloomer and Ives (5) and Komi and Hakkinen (13), this may explain the findings of the training study of Lawton et al. (16), where greater strength gains were found with the CONT loading scheme. It is important to note, however, that the between-subject variability within this study (see *SDs* in Table 3) was greater than that reported by Lawton et al. (16) for concentric time under tension. Further research, with greater subject numbers than this study, is required to further scientific understanding regarding the effect of CONT and ISRV loading on eccentric, concentric, and total time under tension.

It was observed that manipulating the rest period, by decreasing the length of each rest period but increasing the frequency of rest prescribed, did not significantly influence the total kinematic or kinetic responses associated with the CONT and ISRV schemes. It would seem that when the total rest prescribed within a training session is equated, CONT and ISRV schemes may offer a similar mechanical stimulus for the development of strength and power. However, it is important to note that these results do not offer any insight into the effects of decreasing the length of interset rest periods.

It has been proposed that intraset loading may allow the athlete to increase their training volume by training at high intensities for longer durations and performing additional repetitions (4). To study such a contention, a loading scheme that allowed extra repetitions to be performed while equating the rest durations (ISRR) was included. Indeed, it was observed that dividing repetitions and increasing the frequency of the rest periods allowed athletes to perform a greater number of repetitions during the session compared with the CONT loading scheme. These additional repetitions resulted in significantly greater total time under tension, mean force, impulse, mean power, work, and postexercise blood lactate values compared with the other 2 loading schemes.

Explaining the extra repetitions and subsequent superior kinematics and kinetics is difficult. It may be argued that each subject's true 6RM was not established. However, the reliability associated with the 6RM strength assessment was high ($CV = 1.5\%$, $ICC = 0.98$), indicating that the 6RM baseline training load used throughout the testing sessions was stable across testing occasions. This reliability was also similar to values pre-

viously reported for maximal strength assessments (2). Given the reliability of the strength assessment, the training experience of the participants, and the block-randomized experimental design, we consider it unlikely that any of the differences observed between the loading schemes were the result of acute learning effects or strength increases.

It may be speculated that some sort of enhancement occurred similar to that described in the literature that has studied for postactivation potentiation (PAP). PAP is the transient increase in muscle contractile performance after previous contractile activity (18). That is, conditioning activities such as maximal voluntary contractions are thought to enhance the force and/or power output of ensuing contractions/activities (12). The near-maximal voluntary contractions used in this study may have resulted in some form of potentiation that resulted in the extra repetitions, especially early in the loading scheme (e.g., set 2).

More likely, however, is that the increased frequency of the rest periods allowed partial resynthesis of intramuscular phosphocreatine stores, the dominant energy source for heavy resistance training of approximately 5 repetitions (22). It has been previously observed that within 2 minutes of rest, 50% of the depleted intramuscular phosphocreatine stores can be replenished (6). As the rest periods prescribed between sets of ISRR loading were 130 seconds in duration, it would seem possible that similar repletion rates may have occurred in this study. It could therefore be speculated that the increased frequency of the rest periods may have allowed partial resynthesis of the intramuscular phosphocreatine stores, enabling the extra repetitions to be performed. However, the experimental design of this study does not enable further analysis of these possibilities; therefore, the exact cause(s) of the observed enhancement can not be conclusively identified.

Finally, greater postexercise blood lactate accumulation was associated more with the ISRR scheme than the CONT and ISRV schemes. There are a number of mechanisms that may be responsible for such a result. Most likely the greater number of repetitions performed during the ISRR loading scheme and hence greater total work resulted in greater lactate accumulation. It may also be speculated that, as a result of the greater repetition number and associated time under tension, greater vasoconstriction and localized muscle hypoxia occurred, in turn resulting in a greater energy contribution from anaerobic glycolysis. Alternatively, it may be that an accumulative blood lactate effect occurred during the ISRR loading scheme as a result of the short rest periods not allowing sufficient clearance of blood lactate. However, again, the experimental design of this study does not allow further investigation of these possibilities, and the exact cause(s) of the observed increase in lactate accumulation can not be conclusively identified.

PRACTICAL APPLICATIONS

In conclusion, it would seem that intraset rest loading schemes can provide an additional method for manipulating the training variables (e.g., rest, load, volume) to alter the kinematics, kinetics, and resultant hormonal and metabolic responses associated with resistance strength training. If the loading schemes associated with an acute study of this kind are repeated over a number

of occasions, certain strength training responses could be expected given that strength and power adaptation is thought to be influenced by the mechanical, hormonal, and metabolic stimuli associated with various loading parameters (10). Of particular interest are the findings associated with the ISRR loading scheme. The authors take the liberty to briefly speculate on some of the training implications of their findings.

It has been previously suggested that high time under tension (5, 13), mechanical forces, impulses (9), and metabolic accumulation (8, 20) may be important in the development of muscular strength and hypertrophy. Given the significantly greater eccentric, concentric, and total contraction durations, mean forces, impulses, work, and blood lactate responses associated with the ISRR loading scheme, it may be speculated that ISRR loading parameters may offer a superior training stimulus than CONT or ISRV loading parameters for the development of maximal strength and hypertrophy.

In terms of power development, it is thought that training that optimizes the contribution of force and velocity will optimize power development. Although significantly greater total mean power (eccentric, concentric, and overall) was associated with the ISRR loading scheme, from further analysis it seems that the power outputs associated with this scheme were largely based on the higher mechanical forces associated with ISRR loading. Instead, slower movement velocities were associated with ISRR loading. Therefore, it may be speculated that if the development of high power output, irrespective of movement velocity, is the critical stimulus for the development of muscular power, the repeated application of ISRR loading schemes may provide a superior training stimulus to CONT and ISRV loading schemes for the development of muscular power. Alternatively, if movement velocity is important in the development of muscular power, the ISRR loading scheme may not provide an appropriate training stimulus for the development of muscular power. With regard to the CONT and ISRV schemes, similar mean powers were observed between the CONT and ISRV schemes, whereas greater blood lactate levels were associated with CONT loading. Therefore, it may be speculated that ISRV loading may offer a superior training stimulus for the development of muscular power if power training needs to be performed with minimal influence from fatigue (1, 3, 21).

Finally, given that the work was chunked into smaller sets and a greater frequency of rest periods in the ISRV loading scheme, it may be speculated that this scheme resulted in less vasoconstriction of the working muscles than the CONT and ISRR loading schemes. In turn, if the ISRV loading scheme resulted in less vasoconstriction, it may be speculated that this loading scheme may be more appropriate, and safer, for populations susceptible to high blood pressure (hypertension). It may also be speculated that less vasoconstriction may result in less muscle hypoxia/anoxia, potentially minimizing the fast to slow twitch fiber-type transformations previously observed during resistance training (11, 19).

It must be remembered that these results represent group means of the acute kinematic, kinetic, and blood lactate responses to the different loading schemes and should be interpreted with caution. The results of this study do not represent individual responses and do not offer insight into the longitudinal effects of training with

intraset rest loading schemes. However, it would seem that further study into the efficacy of intraset rest loading protocols may be warranted. Such studies may also enhance scientific understanding of rest and its influence on maximal strength and power development. It is recommended that such studies use larger subject numbers and include a wider range of metabolic and hormonal markers than this study.

REFERENCES

1. ABDESSEMED, D., P. DUCHE, C. HAUTIER, G. POUMARAT, AND M. BEDU. Effect of recovery duration on muscular power and blood lactate during the bench press exercise. *Int. J. Sports Med.* 20:368–373. 1999.
2. ABERNETHY, P., G. WILSON, AND P. LOGAN. Strength and power assessment: Issues, controversies and challenges. *Sports Med.* 19:402–417. 1995.
3. BAKER, D. Acute negative effect of a hypertrophy-oriented training bout on subsequent upper-body power output. *J. Strength Cond. Res.* 17:527–530. 2003.
4. BERG, M. Rest-pause for growth: New muscle gain is just a brief pause away with this powerful training technique. *Men's Fitness*. May 2003. Available at: http://www.findarticles.com/p/articles/mi_m1608/is_5_19/ai_100545127. Accessed June 23, 2004.
5. BLOOMER, J. AND J.C. IVES. Varying neural and hypertrophic influences in a strength program. *Strength Cond. J.* 22:30–35. 2000.
6. BOGDANIS, G.C., M.E. NEVILL, H.K.A. LAKOMY, AND L.H. BOOBIS. Power output and muscle metabolism during and following recovery from 10 and 20 s of maximal sprint exercise in humans. *Acta Physiol. Scand.* 163:261–272. 1998.
7. BYRD, R., R. CENTRY, AND D. BOATWRIGHT. Effect of inter-repetition rest intervals in circuit weight training on PWC₁₇₀ during arm-kranking exercise. *J. Sports Med. Phys. Fitness* 28:336–340. 1988.
8. CAREY SMITH, R. AND O.M. RUTHERFORD. The role of metabolites in strength training. 1: A comparison of eccentric and concentric contractions. *Eur. J. Appl. Physiol.* 71:332–336. 1995.
9. CREWTER, B.T. The mechanical, hormonal and metabolic responses to two resistance loading schemes. Master's thesis, Auckland University of Technology, Auckland, 2004.
10. ENOKA, R.M. *Neuromechanics of Human Movement*. Champaign, IL: Human Kinetics, 2002.
11. FRIEDMANN, B., S. BORISCH, K. KUCERA, H. MULLER, G. ERB, G. RICHTER, AND P. BARTSCH. Effects of strength endurance training in normobaric hypoxia on muscle cross sectional area, strength endurance capacity and maximal strength. *Int. J. Sports Med.* 23:S78. 2002.
12. GULLICH, A., AND D. SCHMIDTBLEICHER. MVC-induced short-term potentiation of explosive force. *New Studies Athletics* 4:67–81. 1996.
13. KOMI, P.V., AND K. HAKKINEN. Strength and power. In: *The Olympic Book of Sports Medicine*. A. Dirix, H.G. Knuttgen, and K. Tittel, eds. Boston: Blackwell Scientific, 1988. pp. 181–193.
14. KRAEMER, W.J. A series of studies—The physiological basis for strength training in American Football: Fact over philosophy. *J. Strength Cond. Res.* 11:131–142. 1997.
15. KRAEMER, W.J., B.J. NOBLE, M.J. CLARK, AND B.W. CULVER. Physiologic responses to heavy-resistance exercise with very short rest periods. *Int. J. Sports Med.* 8:247–252. 1987.
16. LAWTON, T., J. CRONIN, E. DRINKWATER, R. LINDSELL, AND D. PYNE. The effect of continuous repetition training and intraset rest training on bench press strength and power. *J. Sports Med. Phys. Fitness* 43:361–368. 2004.
17. LAWTON, T., J. CRONIN, AND R. LINDSELL. Effect of inter-repetition rest intervals on weight training repetition power output. *J. Strength Cond. Res.* 20:172–176. 2006.

18. SALE, D.G. Postactivation potentiation: Role in human performance. *Exerc. Sport Sci. Rev.* 30:138–143. 2002.
19. SALE, D.G., J.D. MACDOUGALL, I. JACOBS, AND S. GARNER. Interaction between concurrent strength and endurance training. *J. Appl. Physiol.* 68:260–270. 1990.
20. TAKARADA, Y., K. SATO, AND N. ISHII. Effects of resistance exercise combined with vascular occlusion on muscle function in athletes. *Eur. J. Appl. Physiol.* 86:308–314. 2002.
21. TESCH, P.A., E.B. COLLIANDER, AND P. KAISER. Muscle metabolism during intense, heavy-resistance exercise. *Eur. J. Appl. Physiol.* 55:362–366. 1986.
22. WEISS, L.W. The obtuse nature of muscular strength: The contribution of rest to its development and expression. *J. Appl. Sport Sci. Res.* 5:219–227. 1991.

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Address correspondence to Jamie Denton, jamie.denton@aut.ac.nz.